

Secure Settings Categorize Options Procedure

Revision

Version 2
4/27/26 6:49 PM

SME

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Abstract

This document describes the procedure used to perform the security option categorization activity described in the **AVCDL** ^[1] secondary document **Secure Settings Document** ^[2].

Group / Owner

Security / Security Architect

Motivation

This document is motivated by the need to ensure that the software is built with tools whose selected settings are based on cybersecurity best practices. This is necessary given the nature of safety-critical, cyber-physical systems.

Note: Within the context of this document, the terms ***security*** and ***cybersecurity*** are used interchangeably. It is presumed that the term ***security*** is being used in reference to ***cybersecurity*** and not ***physical security***.

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Audience

The audience of this document is the cybersecurity practitioner who will be conducting cybersecurity-specific tool setting options categorization.

Completeness of Output

Since the categorization of security options is an as-is analysis, it is critical to ensure that the information gathered is as accurate and complete as possible. As with any cybersecurity activity, it is not the place of the cybersecurity SME to make unsupported assertions. All information should be based on the best available sources and confirmed where possible prior to inclusion.

Note: When information is not available for either a given section of the template or parts thereof, this should be noted.

Disposition of Output

Once completed, the generated output should be managed in the organization's document management system (DMS) as a document of record.

Entry Criteria

This document assumes that the reader understands the purpose of cybersecurity-relevant tool settings. Further, that the reader has read and understood the AVCDL **Secure Settings Document** secondary document.

Prerequisites – Security SME

Qualifications

It is required that the cybersecurity SME is both a qualified and trained security architect (shown above on title page as **Owner**) as defined by the **NIST NCWF** role SP-ARC-002 and detailed in the **NCWF Roles - Security Architect** section of the **AVCDL Background Material** elaboration document [\[5\]](#).

Knowledge

It is required that the cybersecurity SME understands the purpose of cybersecurity-relevant tool settings.

Background Information

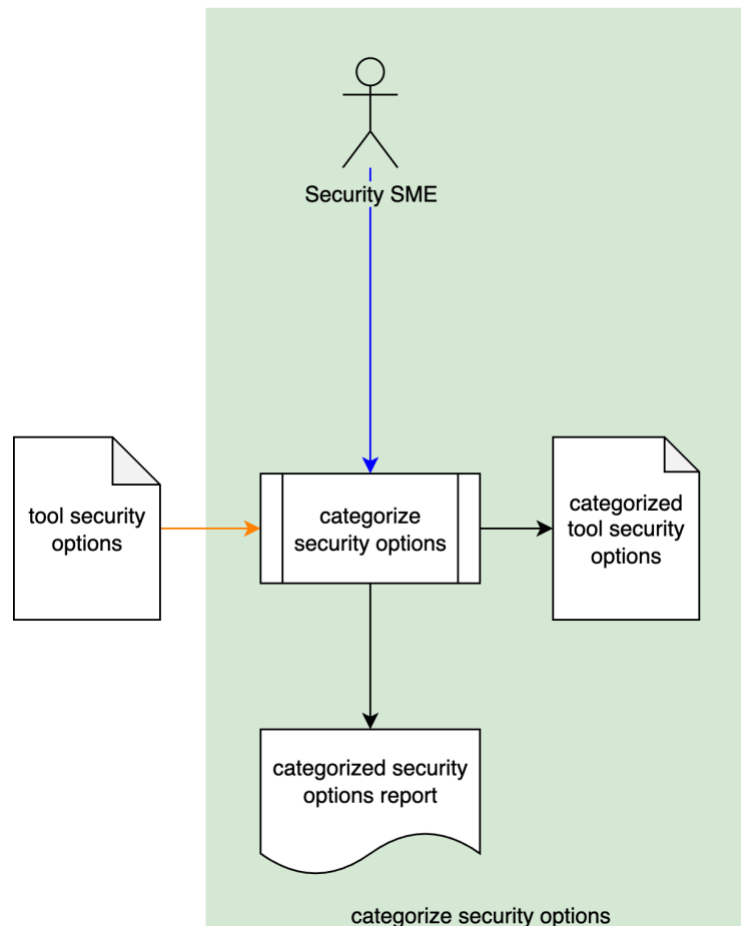
It is required that the cybersecurity SME has read and understands the AVCDL **Secure Settings Document** [\[2\]](#) secondary document. Additionally, that the cybersecurity SME has taken training relevant to this activity.

Prerequisites – Input Materials

It is required that the cybersecurity SME acquire all relevant documentation necessary to complete the procedure.

Categorize Security Options Activity

The workflow diagram for the **categorize security options** activity of the **Secure Settings Document** is shown below.



The Security SME reviews the **Tool Security Options** created in the **Secure Settings – Identify Security Options** ^[4] activity and assigns each a **security priority category**. A **Categorized Tool Security Options** list is generated. A **Categorized Security Options Report** is generated.

Note: It is recommended that the **Categorized Security Options Report** be organized in priority order.

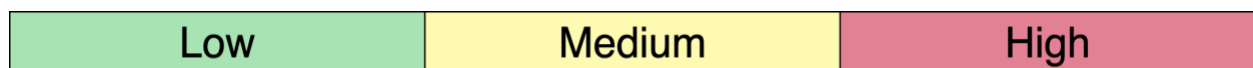
Note: It is important to keep in mind that categorization is done solely from a cybersecurity perspective. This is because not all elements will adopt use of the options based only on category. The categorized options will be further refined during the **Select Security Options** activity.

Security Priority Category

The purpose of assigning a security priority category to each of the options is to allow the development teams the ability to select the options that best fit the organization's priorities. In a perfect world, one could assert that all of the identified cybersecurity-relevant options should be employed in order to achieve the highest level of cybersecurity rigor. This is impractical for all but the most mature (from the product cybersecurity perspective) organizations. As such, it is important to assign a priority to each of the options to facilitate their selection.

Multi-value

When the organization has a pre-existing track record of applying cybersecurity-relevant settings to a particular tool, it may be appropriate to use a multi-value approach, an example of which is shown below.

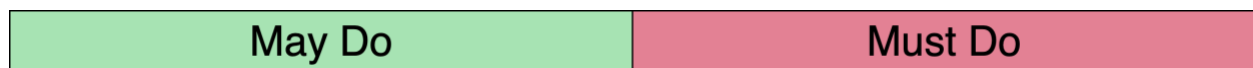


Here we have **low**, **medium**, and **high** values.

It is important when using multi-value categorization, that there be a clear methodology for when options from each category are applied. There is no practical use for highly granular categories if only the most critical one is ever addressed. Keep this in mind when establishing the value set.

Two-value

When the organization is applying cybersecurity-relevant settings to a particular tool for the first time, the best approach is to use a simple two-value categorization as show below.



Here we have only **may do** and **must do** values.

In most situations, this categorization will be sufficient. As the organization becomes increasingly mature with respect to the implementation of the **must do** options, the category criteria may be adjusted and options from the **may do** subset can be reclassified.

Note: Regardless of the category value set, the determinants for each value should documented in such a way that enables any competent cybersecurity SME to apply them consistently.

Methodology

The methodology for this activity is as follows:

1. Define a set of categories.
2. Document the intent of each.
3. Document the qualifications of each.
4. Assign a category to each identified option.

Tool Security Options Template

The tool's cybersecurity-relevant option category information may be documented using the **AVCDL tool cybersecurity options template** Microsoft Excel workbook [\[3\]](#).

Note: Other forms of documentation are permissible so long as they provide the information laid out in this document.

Note: It is presumed that the workbook used was created as a result of undertaking the **Identify Security Options** activity and that only the **categories** sheet and **category** fields in the **options** sheet will need to be entered.

There are six sheets in the workbook. They are:

- [Cover sheet](#)
- [Revision history](#)
- [Reference documents](#)
- [Options](#)
- [Categories](#)
- [Legend](#)

These will be addressed in turn.

Note: This material will include sections previously covered in the **Identify Security Options** activity.

Duplication of Rows in the Various Sheets

When there is the need to add rows to the various sheets of the workbook, be sure to duplicate an existing row. This is because validation checks are attached to some of the cells which also enables the use of dropdown lists.

Cover Sheet

The **cover sheet** of the workbook is shown below:

Tool Cybersecurity Options									
Tool Name	Tool Name								
Tool Version	Tool Version								
Tool Platform	Tool Platform								
Tool URL	Tool URL								
Cybersecurity SME	Cybersecurity SME								
Date	27-Aug-2000								
Revision	1								

Fields to be completed are shown in **red**.

Tool Name

This is the name of the tool whose cybersecurity options are being identified.

Tool Version

This is version number of the tool under consideration.

Tool Platform

This is the platform on which the tool is run.

Tool URL

This is location where the tool can be found.

Cybersecurity SME

This is the cybersecurity subject matter expert creating the options list.

Date

This is the date when the options list was created or updated. The date should be updated whenever the list is updated.

Revision

This is the revision number of the document. The revision number is a monotonic and increasing integer, starting at 1. It should be incremented every time the document is updated.

Revision History

The **revision history** sheet of the workbook is shown below:

Revision History		
Revision	Author	Description
1		initial revision

Revision

The **revision** corresponds to that listed on the cover sheet.

Author

The **author** corresponds to the threat modeling SME listed on the cover sheet.

Description

This is a brief description of changes made to the threat model since it was last updated.

Reference Documents

Note: References relevant to the categorization of the cybersecurity options should be added to this list.

The **references** sheet of the workbook is shown below:

Reference Documents		
Name	Description	Location

Name

This is the name of the document being referenced.

Description

This is a brief description of the document being referenced.

Note: In cases where the reference material is expressed in the form other than a document (for example, a web page), this should be noted.

Location

This is the location of the document being referenced. It may be a physical location or a URL.

Options

The **options** sheet of the workbook is shown below:

Options						
ID	option	description	relevance	bug class	category	notes

ID

This is the unique ID of the option.

Note: An ID of the form **<TOOL>-<zero-leading number>** is recommended. Where multiple tools of the same type (such as compilers) use the same options, a generalized tool type should be used for the ID's prefix.

Option

This is the name of the option.

Description

This is the description of the option.

Relevance

This field describes why the options is relevant from a cybersecurity perspective.

Bug Class

This is the NIST bug class of the option.

Category

This is the option's security priority category.

Note: Values are populated from the **categories** sheet.

Notes

This is a general notes field.

Categories

The **categories** sheet of the workbook is shown below:

Categories				
ID	category	description	qualifications	notes

ID

This is the unique ID of the category.

Note: An ID of the form **<CATEGORY_CLASSIFIER>-<zero-leading number>** is recommended (such as **CAT_LMH-2** or **CAT_BINARY-1**). Where multiple tools of the same categorization use the same values, a generalized categorization methodology name should be used for the ID's prefix.

Category

This is the name of the category.

Description

This is the description of the category.

Qualifications

This field describes why an options would be placed in this category.

Notes

This is a general notes field.

Legend

The **legend** sheet of the workbook is shown below:

NIST bug class
input validation
input verification
memory addressing
memory management
memory use
data declaration
data name resolution
data type computation
data type conversion

The **legend** sheet information is used to make the completion of the document easier by providing dropdown lists for common values. It also ensures that spelling errors do not creep into the generated material.

Note: The legend sheet should not be edited. If an unlisted value is required, the template should be separately revised.

Exit Criteria

This procedure is considered complete once the generated output has been entered into the organization's DMS as a document of record.

Note: The processes and procedures for entering documents into the DMS, or the updating thereof, are outside the scope of this document.

References

1. **AVCDL** (AVCDL primary document)
2. **Secure Settings Document** (AVCDL secondary document)
3. **AVCDL tool cybersecurity options template** (AVCDL template)
4. **Secure Settings – Identify Options Procedure** (AVCDL tertiary document)
5. **AVCDL Background Material** (AVCDL elaboration document)